

Prashant Piprotar

Software Engineer | AI / ML Systems

prashplus@gmail.com | +91 95746 30231 | Bengaluru | www.prashplus.com | linkedin.com/in/prashantpiprotar | github.com/prashplus

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of production experience and an M.Tech in Computer Science from IIT Kanpur, building GPU-accelerated machine learning systems: real-time 3D reconstruction, computer vision and object detection with PyTorch and CUDA, and LLM tooling with LangChain and Model Context Protocol (MCP). Combines distributed systems and HPC expertise with full-stack delivery (FastAPI, Next.js, Docker) to ship ML products end to end.

SKILLS

Languages: JavaScript, TypeScript, Python, Java, C++, SQL
Frameworks & Libraries: React.js, Redux, Next.js, Node.js, Express.js, Spring Boot, FastAPI, Material-UI
Backend & APIs: REST APIs, Microservices, WebSockets, API Design, Model Context Protocol (MCP), LangChain
Databases & Caching: MongoDB, MySQL, Redis
Cloud & DevOps: AWS (EC2, S3), GCP (GKE), Azure (VMs, Blob Storage), Docker, Kubernetes, CI/CD (CircleCI), Git
Distributed Systems & HPC: Apache Spark, OpenMPI, Ceph
AI / ML: PyTorch, CUDA, OpenCV, YOLO, scikit-learn, Computer Vision

PROJECTS

AetherScan 2026
Next.js, FastAPI, PyTorch, Three.js, Docker

- Built a GPU-accelerated 3D reconstruction pipeline with Fast3R and CUDA that converts multi-view images into dense point clouds.
- Streamed point data over WebSockets as raw Float32Array buffers, bypassing React state to render 1M+ points at 60 FPS.
- Containerized the multi-service architecture with Docker Compose and NVIDIA Container Toolkit for reproducible GPU deployments.

Real-Time Object Detection 2025
Python, PyTorch, CUDA, OpenCV, Streamlit

- Developed a real-time object detection service using YOLO with CUDA acceleration for low-latency video inference.
- Built a Streamlit monitoring dashboard with live frame-rate tracking and configurable confidence thresholds for camera streams.

Evaluation of Spark and MPI for Scientific Data Analysis 08/2020 – 05/2021
IIT Kanpur (M.Tech Thesis) | Apache Spark, C++, OpenMPI

- Benchmarked OpenMPI against Apache Spark on distributed clusters, comparing throughput on CPU-, memory-, and I/O-intensive workloads.
- Analyzed performance on large scientific datasets from astrophysics, meteorology, and combustion simulations.

Malware Detection using Machine Learning 01/2020 – 05/2020
IIT Kanpur | Python, scikit-learn

- Trained and evaluated Random Forest classifiers to detect malicious binaries using static and dynamic analysis features.

WORK EXPERIENCE

Senior Software Engineer 06/2023 – Present
Enphase Energy *Bengaluru, India*

- Designed and launched the Solargraf Public REST API and developer documentation in 4 months, reducing partner onboarding time and accelerating third-party integrations.
- Defined engineering standards across the Solargraf platform by improving API performance, expanding automated test coverage, and mentoring engineers across teams.

Software Engineer 08/2021 – 05/2023
Enphase Energy *Bengaluru, India*

- Built an automated Bill of Materials (BOM) calculator with real-time pricing integration in 3 months, increasing customer satisfaction scores by 5%.
- Optimized front-end rendering and back-end database queries across core installer tools, reducing latency and shortening quote turnaround time.

Research Intern 01/2018 – 04/2018
Centre for Development of Advanced Computing (C-DAC) *Pune, India*

- Built an open-source network emulator and integrated IDE for software-defined networking (SDN), reducing setup time for researchers developing SDN applications.
- Deployed a Ceph distributed storage cluster for the SDN Online Lab, improving fault tolerance, high availability, and data retrieval speed.

EDUCATION

M. Tech. in Computer Science and Engineering 07/2019 – 06/2021
Indian Institute of Technology, Kanpur *GPA: 8.6/10*

B. Tech. in Computer Engineering 07/2014 – 05/2018
CSPIT, Charotar University of Science and Technology *GPA: 9.04/10*

XII (C.B.S.E.)
K.V. No. 2, Jamnagar *81%*

X (C.B.S.E.)
K.V. No. 2, Jamnagar *GPA: 10/10*

CERTIFICATIONS

MTA: Windows Server Administration Fundamentals — Microsoft 2017

Elastic Google Cloud Infrastructure: Scaling and Automation; Containers and Services — Google Cloud

Cisco Certified Network Associate: Routing and Switching — Cisco 07/2015 – 04/2017

AWARDS & SCHOLARSHIPS

All India Rank 422, GATE 01/2019